Introduction

Variance measures how far a single random variable ranges around its mean; covariance measures how two random variables move together around their means. Together they populate the covariance matrix of a random vector, which is the central object in multivariate statistics, regression, and portfolio theory.

Prerequisites

Expectation and basic properties of random variables.

Theory

Variance of X:

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

Covariance of X and Y:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

Variance is a special case: $\text{Cov}(X, X) = \text{Var}(X)$.

Properties:

- Non-negativity: $\text{Var}(X) \geq 0$, with equality iff X is constant almost surely.
- Scale: $\text{Var}(aX + b) = a^2\text{Var}(X)$.
- Symmetry: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
- Bilinearity: $\text{Cov}(aX + b, cY + d) = ac\,\text{Cov}(X, Y)$.
- Sum: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.
- Independence implies zero covariance: $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$. The converse is false.

The **covariance matrix** of a random vector $\mathbf{X} = (X_1, \ldots, X_p)$ is the $p \times p$ matrix Σ with entries $\Sigma_{ij} = \text{Cov}(X_i, X_j)$. Σ is symmetric positive semi-definite.

Assumptions

$E[X^2] < \infty$ and $E[Y^2] < \infty$ for variance and covariance to be defined.

R Implementation

```r
library(mvtnorm)
set.seed(2026)

# Univariate variance
X <- rnorm(1e5, mean = 10, sd = 3)
c(theoretical = 9, empirical = var(X))

# Covariance from bivariate normal with known Sigma
Sigma <- matrix(c(4, 1.8,
                  1.8, 9), nrow = 2)
n <- 1e5
XY <- rmvnorm(n, mean = c(0, 0), sigma = Sigma)
cov(XY)
cov(XY) / (sqrt(Sigma[1, 1] * Sigma[2, 2]))   # correlation = 0.3
```

```r
# Variance of a sum verifies the formula
X <- XY[, 1]; Y <- XY[, 2]
c(var_sum = var(X + Y),
  formula = var(X) + var(Y) + 2 * cov(X, Y))

# Zero covariance does not imply independence
X <- rnorm(1e5)
Y <- X^2
c(cov = cov(X, Y),
  correlation = cor(X, Y))
```

Output & Results

```
theoretical   empirical
      9.0         9.04

          [,1]      [,2]
[1,]   3.998     1.791
[2,]   1.791     8.984

          [,1]      [,2]
[1,]   0.999     0.299
[2,]   0.299     0.998

 var_sum formula
  16.56   16.56

       cov   correlation
     0.003         0.002
```

Empirical variance and covariance match the theoretical values within Monte Carlo error. X and X^2 have zero covariance but are clearly dependent – covariance detects only linear association.

Interpretation

Reporting a mean and a variance (or SD) gives a rough summary of a distribution's location and spread; reporting a covariance matrix gives the joint structure of several variables. Most multivariate methods (PCA, MANOVA, linear discriminant analysis) work with sample covariance matrices.

Practical Tips

- Sample variance uses divisor $n-1$ for unbiasedness; R's `var()` uses $n-1$ by default.
- $\mathrm{Cov}(X, Y) = 0$ does not imply independence except in the Gaussian case.
- The covariance matrix must be positive semi-definite; estimation from small samples can produce near-singular matrices.
- Linearly transforming a random vector by matrix A changes the covariance matrix to $A\Sigma A^\top$.
- For robust multivariate covariance estimation in contaminated data, `robustbase::covMcd` is the standard MCD estimator.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots